Juan Manuel Copia

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github.com/JuanmaCopia

Summary

Software engineer and researcher completing a PhD in Computer Science at Universidad Politécnica de Madrid, with hands-on experience at Amazon. I enjoy connecting academic innovation with real-world engineering—combining research in program analysis and automated reasoning with the design and deployment of scalable cloud systems. I'm passionate about the future of artificial intelligence and about building agentic software that's autonomous, adaptive, and impactful.

Experience

Software Development Engineer, *Amazon – Madrid, Spain* May 2025 – Present

- Fast ramp-up into Amazon's large-scale development environment, mastering internal tools, deployment processes, and operational best practices.
- Designed, implemented, and deployed high-impact backend services using **AWS technologies** such as **EC2**, **Lambda**, **S3**, **DynamoDB**, and **CloudWatch**.
- Developed and maintained **infrastructure as code** using **AWS CDK (TypeScript)**, ensuring reproducibility and scalability across environments.
- Automated complex workflows using AI agents and agentic systems
- Contributed to **continuous integration and deployment pipelines**, improving reliability and reducing release cycle time.
- Participated in **peer code reviews**, ensuring code quality, maintainability, and adherence to Amazon's software engineering standards.
- Joined the team's **on-call rotation**, providing 24/7 availability to resolve service incidents and maintain system reliability.
- Mentored new hires as an onboarding buddy, helping them integrate into the team and learn Amazon's development and operational practices.

Research Assistant, *IMDEA Software Institute – Madrid, Spain* Dec 2020 – April 2025

- Designed and implemented a novel Symbolic Execution framework to handle complex heap-allocated inputs. Addressed a key challenge in lazy initialization, improving analysis precision and eliminating false alarms.
- Developed an efficient solver of structural constraints of partially symbolic heaps.
- Developed and applied search-based techniques to automate precondition inference.
- Developed an AI-driven tool that combines Large Language Models (LLMs), random test input generation, and program analysis to improve patch correctness validation. This work integrates ML-based reasoning with automated software testing.

Research Scholarship, *Department of Computer Science – UNRC, Argentina* Oct 2019 – Oct 2020

- Built a Python-based Symbolic Execution engine with support for lazy initialization, enabling automated reasoning for heap-allocated objects.
- Applied the Symbolic Execution engine to generate automated test inputs, successfully identifying a critical bug in an educational open-source data structure.

Summer Internship, *McAfee – Córdoba, Argentina* Jan 2019 – Mar 2019

- Performed fuzz testing on internal software components, uncovering crashes and weaknesses that enhanced the software's robustness and reliability.

Student Teaching Assistant, *Department of Computer Science – UNRC, Argentina* Aug 2018 – Aug 2019

- Supported undergraduate students in 'Data Structures' and 'Algorithms', focusing on algorithm design, implementation, and problem-solving.

Education

PhD in Computer Science, *Universidad Politécnica de Madrid – Madrid, Spain* Oct 2021 – Present

Undergraduate degree in Computer Science (5 years + thesis) Mar 2015 – Dec 2020
Department of Computer Science, UNRC, Argentina

- GPA: 9.02/10.0

Undergraduate degree in Computer Science (3 years + final project) Mar 2015 – Jun 2018
Department of Computer Science, UNRC, Argentina

- GPA: 8.81/10.0

Publications

Search-based Inference of Class Invariants: How far can Simulated Annealing take us? Coming in Nov 2025

J. M. Copia, F. Molina, N. Aguirre, M. Frias, A. Gorla, P. Ponzio.
To appear in Symposium on Search-Based Software Engineering (SSBSE) 2025, Seoul, Republic of Korea

Search-based Inference of Class Invariants (Poster) Aug 2025

J. M. Copia, F. Molina, N. Aguirre, M. Frias, A. Gorla, P. Ponzio.
2025 Genetic and Evolutionary Computation Conference (GECCO), Malaga, Spain
doi: 10.1145/3712255.3726698

Improving Patch Correctness Analysis via Random Testing and Large Language Models May 2024

F. Molina, ***J. M. Copia***, A. Gorla.
2024 IEEE Conference on Software Testing, Verification and Validation (ICST), Toronto, ON, Canada, 2024, pp. 317-328.
doi: 10.1109/ICST60714.2024.00036

Precise Lazy Initialization for Programs with Complex Heap Inputs Oct 2023

J. M. Copia, F. Molina, N. Aguirre, M. Frias, A. Gorla, P. Ponzio.
2023 IEEE 34th International Symposium on Software Reliability Engineering (ISSRE), Florence, Italy, 2023, pp. 752-762.
doi: 10.1109/ISSRE59848.2023.00080

LISSA: Lazy Initialization with Specialized Solver Aid Oct 2022

J. M. Copia, P. Ponzio, N. Aguirre, A. Gorla, M. Frias.
37th IEEE/ACM International Conference on Automated Software Engineering (ASE '22). Association for Computing Machinery, New York, NY, USA, Article 67, 1–12.
doi: 10.1145/3551349.3556965

Use of Test Doubles in Android Testing: An In-Depth Investigation May 2022

M. Fazzini, C. Choi, ***J. M. Copia***, G. Lee, Y. Kakehi, A. Gorla, A. Orso.
2022 IEEE/ACM 44th International Conference on Software Engineering (ICSE), Pittsburgh, PA, USA, 2022, pp. 2266-2278.
doi: 10.1145/3510003.3510175

Research Prototypes

- **Express**: A tool that automatically infers method preconditions and class invariants using search-based algorithms. It specializes in generating executable predicates (Java code) as preconditions for complex heap-allocated objects.

  Available at: github.com/JuanmaCopia/express

- **FixCheck**: A tool for improving patch correctness analysis in Java. It integrates static analysis, random testing, and large language models (LLMs) to automatically generate tests that highlight and explain potential patch inconsistencies.

  Available at: github.com/facumolina/fixcheck

- **PLI**: A Symbolic Execution framework built on top of Java Symbolic PathFinder (JPF). It enhances symbolic analysis for programs with complex heap-allocated inputs and intricate preconditions. PLI addresses key challenges in lazy initialization, enabling more precise and efficient symbolic reasoning.

  Available at: github.com/JuanmaCopia/spf-pli

- **SymSolve**: An efficient bounded exhaustive solver for symbolic structures with complex representation invariants. It determines the satisfiability of partially symbolic heaps with respect to structural constraints by leveraging operational specifications (e.g., repOk routines).

  Available at: github.com/JuanmaCopia/SymSolve

- **PySEAT**: A Symbolic Execution engine for Python programs that implements lazy initialization to represent heap-allocated objects symbolically. It automates test generation for Python codebases, achieving high coverage and uncovering potential errors in complex systems.

  Available at: github.com/JuanmaCopia/PySEAT

Public Talks

Precise Lazy Initialization for Programs with Complex Heap Inputs Apr 2024
Workshop, KLEE Workshop on Symbolic Execution, *Lisbon, Portugal.*

Precise Lazy Initialization for Programs with Complex Heap Inputs Oct 2023
Research Track, ISSRE 2023, *Florence, Italy.*

LISSA: Lazy Initialization with Specialized Solver Aid Oct 2022
Research Track, ASE 2022, *Oakland Center, Michigan, USA.*

LISSA: Lazy Initialization with Specialized Solver Aid Sep 2022
Oral communication, IMDEA Software S3 Seminar Series, *Madrid, Spain.*

A Satisfiability Solver for Symbolic Structures with Complex Representation Invariants Mar 2022
Oral communication, FACAS 2022, *La Falda, Córdoba, Argentina.*

Technical Skills

Programming: Python, Java, TypeScript
Artificial Intelligence: OpenAI Agents SDK, MCPs, PyTorch, scikit-learn, NumPy, pandas, Matplotlib, data preprocessing, model evaluation
AWS: EC2, Lambda, S3, DynamoDB, SQS, SNS, CloudWatch, IAM, CloudFormation, CDK (Infrastructure as Code)
Software Engineering: Unit Testing, Integration Testing, Code Review
Java Frameworks and Libraries: Dagger, Lombok, JUnit
Research and Analysis: Static and dynamic analysis, automatic test input generation, symbolic execution, fuzzing, invariant inference
Tools: Git, Docker

Spoken Languages: English (fluent), Spanish (native), French (fluent)